

Environmental toxicants and the developing immune system: a missing link in the global battle against infectious disease?

There is now compelling evidence that developmental exposure to chemicals from our environment contributes to disease later in life, with animal models supporting this concept in reproductive, metabolic, and neurodegenerative diseases. In contrast, data regarding how developmental exposures impact the susceptibility of the immune system to functional alterations later in life are surprisingly scant. Given that the immune system forms an integrated network that detects and destroys invading pathogens and cancer cells, it provides the body's first line of defense. Thus, the consequences of early-life exposures that reduce immune function are profound. This review summarizes available data for pollutants such as cigarette smoke and dioxin-like compounds, which consistently support the idea that developmental exposures critically impact the immune system. These findings suggest that exposure to common chemicals from our daily environment represent overlooked contributors to the fact that infectious diseases remain among the top five causes of death worldwide.